Reg.No.									

INTERNATIONAL CENTRE FOR APPLIED SCIENCES (MAHE)

III SEMESTER B.Sc.(Applied Sciences) DEGREE EXAMINATION – NOV 2021

SUBJECT: MATERIAL SCIENCE AND METALLURGY (IME 233)

(BRANCH: MECHANICAL)

Time: 3 Hours **DATE: 24 NOV 2021** **Max. Marks: 50**

- ✓ **Answer ANY FIVE full Questions.**
- ✓ **Missing data, if any, may be suitably assumed**

1A Draw the neat sketch of FCC cubic unit cell and find i) lattice parameter in terms of atomic radius ii) co-ordination number iii) effective number of atoms per unit cell

1B Compare edge and screw dislocations with the help of neat sketches along with their Burger's circuit.

1C Show the following planes and directions in cubic systems: $(\bar{2}00)$, (100), $[\bar{2}10]$, $[121]$

(3+5+2=10 Marks)

2A What are solid solutions? Explain different types of solid solutions with necessary sketches.

2B Explain the mechanism of solidification of liquid metals.

2C How solidification of an alloy is different from pure metal?

(3+5+2=10 Marks)

3A Discuss the application of the lever rule in a phase diagram.

3B Write a short note on intermetallic and interstitial compounds with examples.

3C Explain construction method of Isomorphous phase diagram

(2+3+4=10 Marks)

4A Melting temperatures of Copper (Cu) and Silver (Ag) are 1080^0C and 960^0C respectively. The metals Copper and Silver are mutually soluble in the liquid state and partly soluble in the solid state. A liquid phase alloy containing 70% Silver completely transforms into a mixture of two solid solutions at 780^0C. The maximum solubility of Ag in Cu and Cu in Ag are 8% and 10% respectively at 780^0C. The room temperature solubility is negligible. Assuming the curves to be linear, draw a phase diagram to scale and label the regions. For 40% Ag alloy, determine the following:
i)Composition of the nucleus. ii)Composition of the last drop of the liquid precipitating iii) Weight percentage of eutectic formed at 780^0C. iv)Weight ratio of the solid phases in the eutectic mixture v) Composition of the liquid phase undergoing eutectic reaction

4B Draw an Iron -Carbon phase diagram and label the regions. Write the invariant reactions involved in it. Also draw the cooling diagram of pure iron.

(5+5 Marks)

5A Write the construction procedure of the TTT diagram for 0.8% carbon steel and superimpose the cooling curves to obtain martensitic and bainitic structure.

5B Write composition, properties and applications of various plain carbon steels

5C Distinguish between, i) Annealing and Normalising ii) Hardness and Hardenability

(5+3+2 Marks)

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